

ASSIGNMENT -3.4

1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home Wi-Fi router, data center, college campus backbone).

Answer: -

Ethernet Cable vs Fiber Optic Cable

Feature	Ethernet Cable	Fiber Optic Cable
1. Transmission	Uses electrical signals through copper wires.	Uses light signals through glass/plastic fibers.
2. Speed & Distance	Good speed, but generally shorter transmission distance.	Very high speed and can transmit over much longer distances.
3. Cost & Installation	Cheaper and easier to install.	More expensive and requires specialized equipment/installation.

Real-world examples

- **Ethernet:** Connecting a **home Wi-Fi router to a computer, printer, or switch.**
- **Fiber optic:** Connecting **different buildings on a college campus or high-speed data-center networks.**

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

Answer:-



Pin	Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Answer: -

Step-by-Step Guide: Attaching an RJ45 Connector to a CAT6 Cable

Tools required: CAT6 cable, RJ45 connector, RJ45 crimping tool, wire stripper/cutter, and optionally a LAN cable tester.

1. **Cut the CAT6 cable** to the required length.
2. **Strip about 2–3 cm of the outer jacket** carefully without damaging the inner wires.
3. **Separate and straighten the 8 wires** by gently untwisting the four pairs.
4. **Arrange the wires in T568B order:**
 - White/Orange
 - Orange
 - White/Green
 - Blue
 - White/Blue
 - Green
 - White/Brown
 - Brown
5. **Trim all 8 wires evenly** so their ends are straight and approximately the same length.
6. **Insert the wires into the RJ45 connector**, keeping the T568B order. Push them fully to the front of the connector.
7. **Check the wiring** through the transparent connector before crimping.
8. **Place the RJ45 connector into the crimping tool** and squeeze the handles firmly until the connector is securely crimped.
9. **Inspect the connector** to make sure the cable jacket is held securely and all wires are properly seated.
10. **Test the cable** with a LAN cable tester. A correctly wired cable should show the pins in the proper sequence.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.

Answer: -

For a gaming café with 10 PCs, I would choose **Ethernet (CAT6) cables** because they provide enough speed for LAN gaming and are more affordable for short distances. I would use an **RJ45 crimper** to attach the connectors to the CAT6 cables. Fiber optics would be faster over long distances, but it would be more expensive and require specialized equipment. Ethernet is therefore the better choice for a small gaming café because it is **fast, reliable, easy to install, and budget-friendly**.